INCH-POUND
MIL-H-2217D(SH)
9 July 1993
SUPERSEDING
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7 July 1976

MILITARY SPECIFICATION

HOSE ASSEMBLIES, WIRE-REINFORCED RUBBER, SUBMARINE RESCUE CHAMBER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 1- and 1-1/4-inch inside diameter (id), wire-reinforced hose assemblies for use on submarine rescue chambers designed to operate at depths up to 850 feet.

1.2 Classification. Hoses are of the following sizes and lengths, as specified (see 1.2.1, 3.4 and 6.2).

1.2.1 Hose assembly size. Hose assembly size is identified by a letter code as shown in table I (see 6.9).

TABLE I. Size designation.

Code letter	Size (id) (inches)
A	1
B	1-1/4

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 05Q42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Hose assembly length. Hose assembly length is identified by a two-digit number directly indicating length in feet (see 3.4 and 6.9).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
TT-S-735 - Standard Test Fluids, Hydrocarbon.

MILITARY

MIL-H-775 - Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing) and Associated Hardware: Packaging of.

STANDARDS

FEDERAL

FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.
FED-STD-601 - Rubber: Sampling and Testing.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 380 - Standard Test Methods for Rubber Hose. (DoD adopted)
D 750 - Standard Test Method for Rubber Deterioration Carbon - Arc Weathering Apparatus.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Material. Material shall be new and unused. The hose shall have been manufactured not greater than 1 year prior to date of delivery.

3.2.1 Inner tube. The basic rubber stock of which the inner tube is fabricated shall be natural rubber or a synthetic rubber.

3.2.2 Cover. The basic rubber stock of which the cover is fabricated shall be polymerized chloroprene.

3.2.3 Wire reinforcement. The reinforcing helix wire or wires shall be heavily zinc-coated (galvanized) or copperized, round steel wire(s) of a diameter and spacing to meet the internal and external hydrostatic pressure requirements specified in 3.4.2, 4.3, table VI and 4.6.1.

3.2.4 Couplings. Each length of hose shall be fitted with a male coupling on one end and a female coupling on the other end. The female coupling shall be fitted with a washer. The hose end of the couplings shall be so designed as to grip the hose from both the inside by expanding the nipple and outside by swaging the collar. The male and female threaded portion of the couplings shall conform to figures 1 and 2 and the swivel nut to figure 3. The collars shall conform to figure 4. Each hose assembly shall be provided with one set of hexagonal clamps conforming to figure 5.

3.2.5 Organic solvents. The organic solvents used at any stage in the construction of the hose, including the solvents used for rubber compounds for fastening couplings to the hose, for sealing or capping the ends of the hose, for bonding the layers and plies of the air or gas mixture stream passing through the hose, shall conform to the requirements specified in 3.2.5.1.

3.2.5.1 Composition. Only solvents having a threshold limit value (TLV) of not greater than 500 milligrams per cubic meter (mg/m^3) shall be used (see 6.3 and 6.5).

3.2.6 Other rubber additives. The hose shall contain the minimum quantities necessary of only the purest grades of rubber accelerators, both in the cover and inner tube so as to be as free from odor as possible. The following ingredients shall not be used: pine tar, coal tar, benzidine, beta-naphthylamine and derivatives, phosphate plasticizers, and aniline (see 6.3 and 6.5).

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3.2.7 Coupling lubricants and washer dressing. Coupling lubricants or washer dressings shall be used.

3.3 Construction.

3.3.1 Hose. The hose assembly shall consist of an inner rubber tube, two or more plies of fabric braid reinforcement with a rubber ply between the braids, a wire helix or helixes running the entire length of the hose. Each end of the hose shall be capped, or sealed, with a 1/16 inch thick rubber compound similar to the tube, to prevent water contact with either the braid or helix wire.

3.3.1.1 Synthetic rubber filling strip. The space between the adjacent wires shall be filled with strips of a compatible rubber compound as is used in the caps. The strips shall be of such thickness that the outside of the hose will not show appreciable corrugations.

3.3.2 Couplings. Each length of hose shall be fitted with a male coupling on one end and a female coupling on the other end. The female coupling shall be fitted with a washer. The hose end of the coupling shall be so designed as to grip the hose from both the inside by expanding the nipple and outside by swaging the collar. The hose fitting components shall be as specified in table II.

TABLE II. Hose fitting components.

Nipple	Figure 1
Coupling	Figure 2
Union nut	Figure 3
Collar	Figure 4
Coupling retainer	Figure 5

3.3.3 Clamps. Each hose assembly shall include one hexagonal retainer clamp. The clamp shall conform to figure 5.

3.4 Physical requirements. Hose and hose assemblies shall meet the physical requirements of table III. Unless otherwise specified (see 6.2), the hose assembly length shall be 50 feet nominal.

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TABLE III Physical requirements.

Requirement	Intake (1 inch) (id)	Vent 1-1/4 inch (id)	Requirement	Test
Size Inside diameter (id) (inches) Outside diameter (od) (inches) Length, nominal (feet)	1 + 1/32, -0 1-27/32 + 1/8, -1/16 50	1-1/4 + 1/32, -0 2-1/8 + 1/8, -1/16 50	3.4.1	4.5
Thickness (minimum) Tube (inch) Cover (inch)	5/64 3/64	5/64 3/64	3.4.1	4.5
Hydrostatic pressure (minimum) External proof (psi) Internal proof (psi) Internal proof (psi) (1 minute)	600 600 725	600 600 725	3.4.2	4.6.1
Adhesion (minimum) Tube to carcass lbf Cover to carcass lbf	13 13	13 13	3.4.3	4.6.2
Tensile strength (minimum) Tube (psi) Cover (psi) Hose assembly	1600 1800	1600 1800	3.4.4 3.7	4.6.3
Ultimate elongation (percent) Tube and cover	300	300	3.4.5	4.6.3
Volume swell cover (percent) (maximum)	50	50	3.4.6	4.6.4

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3.4.1 Dimensions. The hose OD, ID, and lengths shall conform to table III when measured as specified (see 4.5). Tube and cover thickness shall conform to table III.

3.4.2 Hydrostatic pressure.

3.4.2.1 Internal proof and burst. The hose assemblies shall withstand the proof and burst test pressures specified in table III when tested as specified (see 4.6.1.1 and 4.6.1.2). There shall be no leakage through the hose, hose fitting, or around the junction of the hose and fitting. There shall be no evidence of coupling slippage.

3.4.2.2 External hydrostatic compression. The hose shall withstand the external pressure specified in table III when tested as specified (see 4.6.1.3). During this test a ball shall be pulled through the hose. The force to pull the ball through the hose shall be not greater than 30 pounds. The ball size shall be as specified in table IV.

TABLE IV. Ball size.

Hose size	Ball size (inch)
A	0.650
B	0.812

3.4.3 Adhesion between plies. The adhesion between plies shall meet the requirements of table III when tested as specified (see 4.6.2).

3.4.4 Tensile strength, tube and cover. The initial tensile strength of the tube and cover shall comply with the values of table III when tested as specified (see 4.6.3). Tensile strengths after the immersion of 4.6.3.1 shall not show a decrease of greater than 50 percent. Tensile strengths after aging (see 4.6.3) shall not show a decrease from table III values of greater than 25 percent unless the test specimen thickness is less than 0.05 inch in which case a maximum decrease of 35 percent will be permitted. Tensile strength after light aging (see 4.6.3) shall retain 75 percent of the initial value specified in table III.

3.4.5 Elongation. The elongation shall comply with the value specified in table III when tested as specified (see 4.6.3).

3.4.6 Volume swell. The volume swell shall comply with the value specified in table III when tested as specified (see 4.6.4).

3.5 Bend. There shall be no evidence of cracking, ply separation, or deformation of wire helix when tested as specified (see 4.6.5).

3.6 Off-gassing. The hose shall not contaminate the air passing through it more than that allowed in table V when tested as specified (see 4.6.6).

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TABLE V. Maximum allowable concentration of hydrocarbons in air passing through hose.

Hose length (feet)	Hydrocarbon concentration as methane equivalent (mg/m^3)
3-3/4	4
50	100

3.7 Tensile strengths of hose assembly. The hose assembly shall not fail at less than 2400 pounds for the 1-inch and 3000 pounds for the 1-1/4 inch when tested as specified (see 4.6.7). Failure shall be in the hose and shall not be evidenced by coupling pull off.

3.8 Branding. Each hose shall be permanently marked with a continuous, embossed, layline containing the following in letters not less than 1/4 inch high:

(a) MIL-H-2217.
(b) Manufacturer's name, logo, or both.
(c) 600 pounds per square inch (lb/in^2).
(d) SRC.
(e) Quarter and year manufacturer.

3.9 End protection. After fabrication and testing, each end shall be capped, plugged, or taped to protect the threads and prevent entry of foreign matter into the hose assembly.

3.10 Workmanship. Hose assemblies shall be free of dirt, oil, and metal particles. Swaged pieces shall be free from swage draw marks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

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contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on one hose assembly of each size. The inspection shall include the examinations and tests specified in table VI.

TABLE VI. First article and quality conformance inspections.

Test	Requirement	Test method	First article	Quality conformance
Visual and dimensional examination	3.4.1	4.5	X	X
Hydrostatic				
Proof	3.4.2.1	4.6.1.1	X	See 4.4.2
Burst	3.4.2.1	4.6.1.2	X	X
External proof	3.4.2.2	4.6.1.3	X	See 4.4.2
Adhesion between plies	3.4.3	4.6.2	X	
Tensile				
Tube	3.4.4	4.6.3	X	
Cover	3.4.4	4.6.3	X	
Elongation	3.4.5	4.6.3	X	
Volume swell	3.4.6	4.6.4	X	
Bend	3.5	4.6.5	X	
Off-gassing	3.6	4.6.6	X	X
Hose assembly tensile	3.7	4.6.7	X	X
Branding	3.8			
End protection	3.9			
Workmanship	3.10			

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in table VI (see 6.3 and 6.7).

4.4.1 Inspection lot. For the purpose of quality conformance inspection and test sampling, a lot is defined as all the hose lengths, with couplings, of the same size and type, produced in one facility, using the same production processes and materials, being offered for delivery at one time.

4.4.2 Sampling for hydrostatic proof tests. Each hose assembly shall be subjected to the hydrostatic internal and external proof tests as specified in 4.6.1.1 and 4.6.1.3.

4.4.3 Sampling for visual and dimensional examination. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed hose lengths in accordance with table VI. If one or more defects are found in any sample, the entire lot shall be rejected. The contractor has the option of screening 100% of the lot for the defective characteristics or providing a now lot which shall be inspected in accordance with the sampling plan contained herein. The contractor shall maintain for a period of 3 years after contract completion all records of inspections, tests and any resulting rejections.

TABLE VII. Sampling for visual and dimensional examination.

Lot size	Sample size	
	Major defects	Minor defects
2 to 15	13	2
16 to 25	13	3
26 to 90	13	5
91 to 150	13	6
151 to 280	20	7
281 to 500	29	9
501 to 1,200	34	11
1,201 to 3,200	42	13
3,201 to 10,000	50	15

4.4.4 Sampling for quality conformance tests. Except for hydrostatic proof tests (see 4.4.2), sample hose of sufficient length for testing shall be taken from each lot in accordance with table VIII and subjected to the tests as specified in table VI. Selected hoses shall be fabricated in lengths longer than required so that after removal of the hose test sample the resulting hose assembly shall meet the length requirement of 3.4.1.

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TABLE VIII. Sampling for quality conformance tests.

Lot size	Sample size
2 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1,200	19
1,201 to 3,200	23
3,201 to 10,0000	29

4.4.5 Lot rejection. If one or more defects are found during any examination or test the hose shall be rejected, and the lot represented by the sample shall be rejected. Rejection of a lot shall require that corrective action be implemented by the contractor.

4.5 Visual and dimensional examination. The sample lengths of hose selected as specified in 4.4.3 shall be examined to determine conformance to the dimensional requirements of this specification. The samples shall also be examined for visual defects. FED-STD-162 shall be used to determine and evaluate visual defects.

4.6 Test methods.

4.6.1 Hydrostatic tests.

4.6.1.1 Internal proof. Each length of hose shall be subjected to the internal proof pressure as specified in 3.4.2.1 and table III (see 6.3 and 6.6). The pressure shall be held for 5 minutes. Test procedure shall be in accordance with ASTM D 380. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.1.2 Internal burst. Each designated sample shall be subjected to the internal burst pressure as specified in 3.4.2.1 and table III. Test procedure shall be in accordance with ASTM D 380. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.1.3 External pressure. Each hose assembly shall be subjected to the external pressure as specified in 3.4.2.2 and table III (see 6.3 and 6.6). The test chamber for this test shall be similar to that shown on figure 6. During this test a brass ball shall be drawn through the hose using a force of not greater than 30 pounds. Size of the brass ball shall conform to table IV. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.2 Adhesion tests. Adhesion of the various plies shall be determined in accordance with ASTM D 380 static mass method, type A specimens. Nonconformance to 3.4.3 shall constitute failure of this test.

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4.6.3 Tensile strength and elongation test. Tensile strength and elongation of the designated samples shall be determined in accordance with ASTM D 380. Nonconformance to 3.4.4 or 3.4.5 shall constitute failure of this test. Prior to conducting this test the sample shall have been aged in accordance with ASTM D 750.

4.6.3.1 Immersion. Compliance with 3.4.4 shall be determined by method 6111 of FED-STD-601 using type I fluid in accordance with TT-S-735. Tensile strength shall be based on the swollen area.

4.6.4 Volume swell. Volume swell shall be determined in accordance with ASTM D 380. Nonconformance to 3.4.6 shall constitute failure of this test.

4.6.5 Bend test. A sample hose, not less than 24 inches long shall be bent 180 degrees around an 8-inch diameter mandrel to ensure compliance with the requirement specified in 3.5. The hose shall also withstand being coiled on a 2-foot diameter reel without permanent set. Noncompliance with 3.5 shall constitute failure of this test.

4.6.6 Off-gassing. Measurements shall be made of the off-gassing of organic compounds from the hose. A Beckman Total Hydrocarbon Analyzer, Model 400, or any equivalent process analyzer employing a hydrogen Flame Ionization Detector (FID) having a range from 0 to at least 1,000 mg/m^3 methane equivalents (see 4.6.6.1) shall be used. The measurements shall be made only on hoses which have not been flushed with air, gas, or water, or which have been stored at a temperature not less than 23 degrees Celsius (°C) for a period of at least 7 days since the hoses were flushed. During the 7-day storage period, the ends of the hoses shall either be sealed shut or fastened together so that the confined gas may not escape. Both the analyzer and hoses shall be maintained at a temperature not less than 23°C throughout the testing period. The procedure for measuring off-gassing rates from hoses is specified as the flush test (see 4.6.6.2). Noncompliance with 3.6 shall constitute failure of this test.

4.6.6.1 Terminology. The FID of the Total Hydrocarbon Analyzer is sensitive not only to hydrocarbons themselves, but also to related compounds which will ionize in a hydrogen flame. The term hydrocarbon is intended to designate all organic compounds detected by the Total Hydrocarbon Analyzer. The term methane equivalents designates the concentration of methane that will cause the instrument to give the same reading as the air or other gas tested. All units are expressed in terms of measurements made at 23°C and at a pressure of 760 millimeter of mercury (mmHg).

4.6.6.2 Flush test. By this procedure, measurements shall be made of the increase in the hydrocarbon concentration of a stream of air flowing through the hose at a velocity of 28 liters per minute (1 cubic foot per minute). The temperatures of the hose, air supply, and analyzer shall be not less than 23°C. The flow arrangement shall be as specified on figure 7. Before the air passes through the hose, the air shall be clean and shall contain not more than 1 mg/m^3 of hydrocarbons (methane equivalents). The analyzer shall be zeroed with air passing at the stipulated velocity and temperature through the connector tubes only. The hose shall then be inserted in the line and the air stream passed through it. For the ensuing 15 minutes, readings of the hydrocarbon

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d in 4.6.9. to be used in performing the hydraulic impulse test specifi

concentration shall be taken continuously. The hoses shall be rated on the reading at the end of the 15-minute test period. Hoses which contaminate the air by amounts greater than as specified in 3.6 shall not be acceptable.

4.6.7 Hose assembly tensile test. The designated hose assemblies shall be subjected to a tensile pull to failure as specified in 3.7. Noncompliance with 3.7 shall constitute failure of this test.

4.7 Inspection of packaging. Sample packages, and the inspection of packaging (preservation, packing and markings) for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging requirements. The packaging (preservation, packing and marking) requirements shall be in accordance with MIL-H-775 for the level of preservation (A, C or Commercial), the level of packing (A, B, C, or Commercial), and marking including other packaging acquisitioning options therein, as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assemblies covered by this specification are intended to carry air and other breathing gas mixtures to submarine rescue chambers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Size and length, if other than as specified (see 1.2 and 3.4).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Whether first article inspection is required (see 3.1).
(e) Levels of preservation, level of packing and other options required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific

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acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.5.1, 3.2.6, 4.6.1.1, and 4.6.1.3	DI-MISC-80678	Certification/data report	10.3.1 does not apply
4.4	DI-T-2072	Reports, test	-----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Material certification. The manufacturer should certify to the solvent TLV's and that prohibited rubber additives have not been used (see 3.2.5.1 and 3.2.6).

6.6 Hydrostatic test certification. Each hose assembly should be accompanied by a certificate of satisfactory internal proof and external pressure tests (see 4.6.1.1 and 4.6.1.3). Each certificate should contain the signature or symbol of the contractor's inspector.

6.7 Quality conformance inspection reports. Each lot should be accompanied by a satisfactory quality conformance inspection report and satisfactory report sheets for all designated samples (see 4.4). Each report should contain the signature or symbol of the contractor's inspector and the signature or symbol of the Defense Contract Administration Services Management Area (DCASMA) representative.

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6.8 Lot acceptance and rejection criteria.

6.8.1 Tests. Any length of hose in the sample failing one or more tests should be rejected. The lot represented by the sample should be rejected.

6.9 Part or identifying number (PIN). The PIN of the hose assemblies covered by this specification should consist of the letter "M", the basic number of this specification, and coded numbers or letters as shown in the following example:

Hose assembly

M 2217 - A - 50 Example of PIN: M2217-A-50

M prefix - Military specification

Basic specification number

Size (id) (see 1.2.1 and table I)

Length (see 1.2.2)

6.10 Subject term (key word) listing.

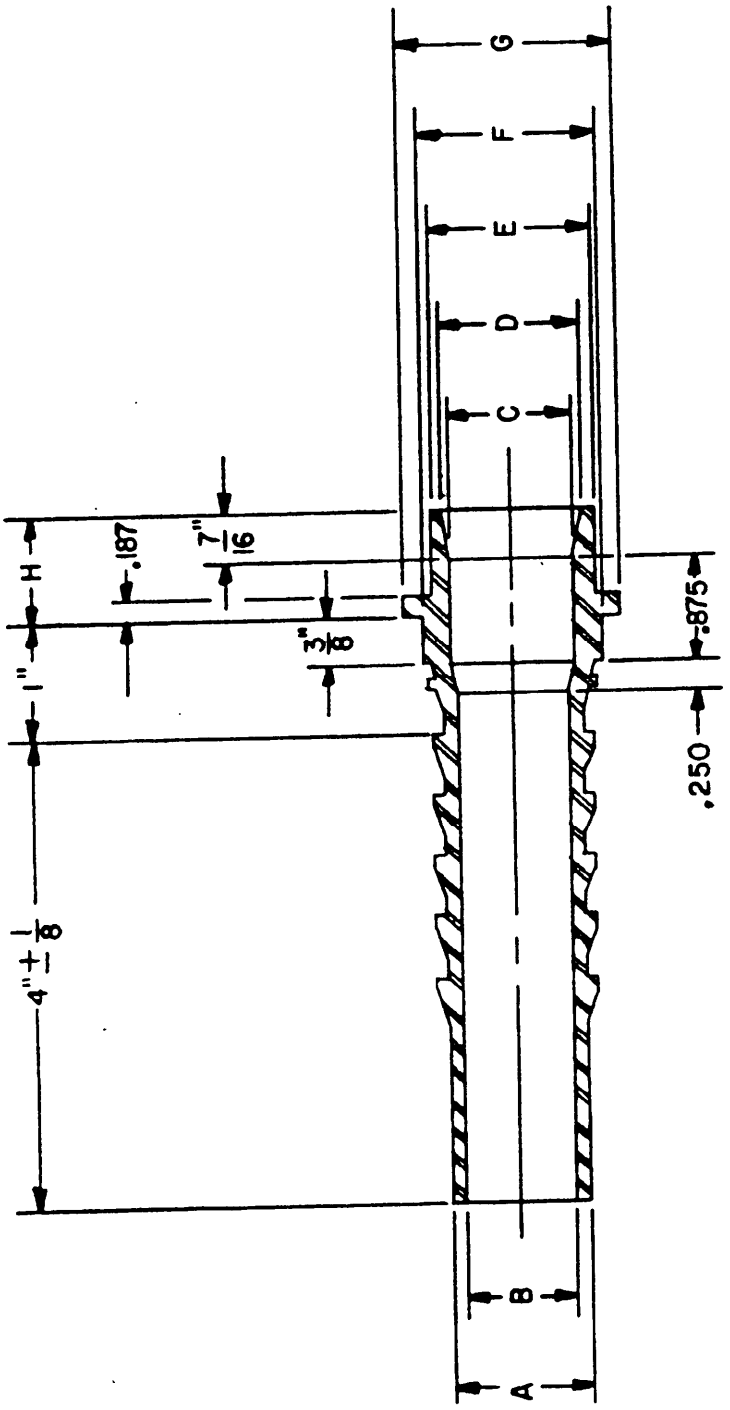
Breathing gas mixture
Hexagonal retainer clamp
Hose coupling
Hydrostatic test
Polymerized chloroprene

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Review activity:
DLA - CS

Preparing activity:
Navy - SH
(Project 4720-N051)

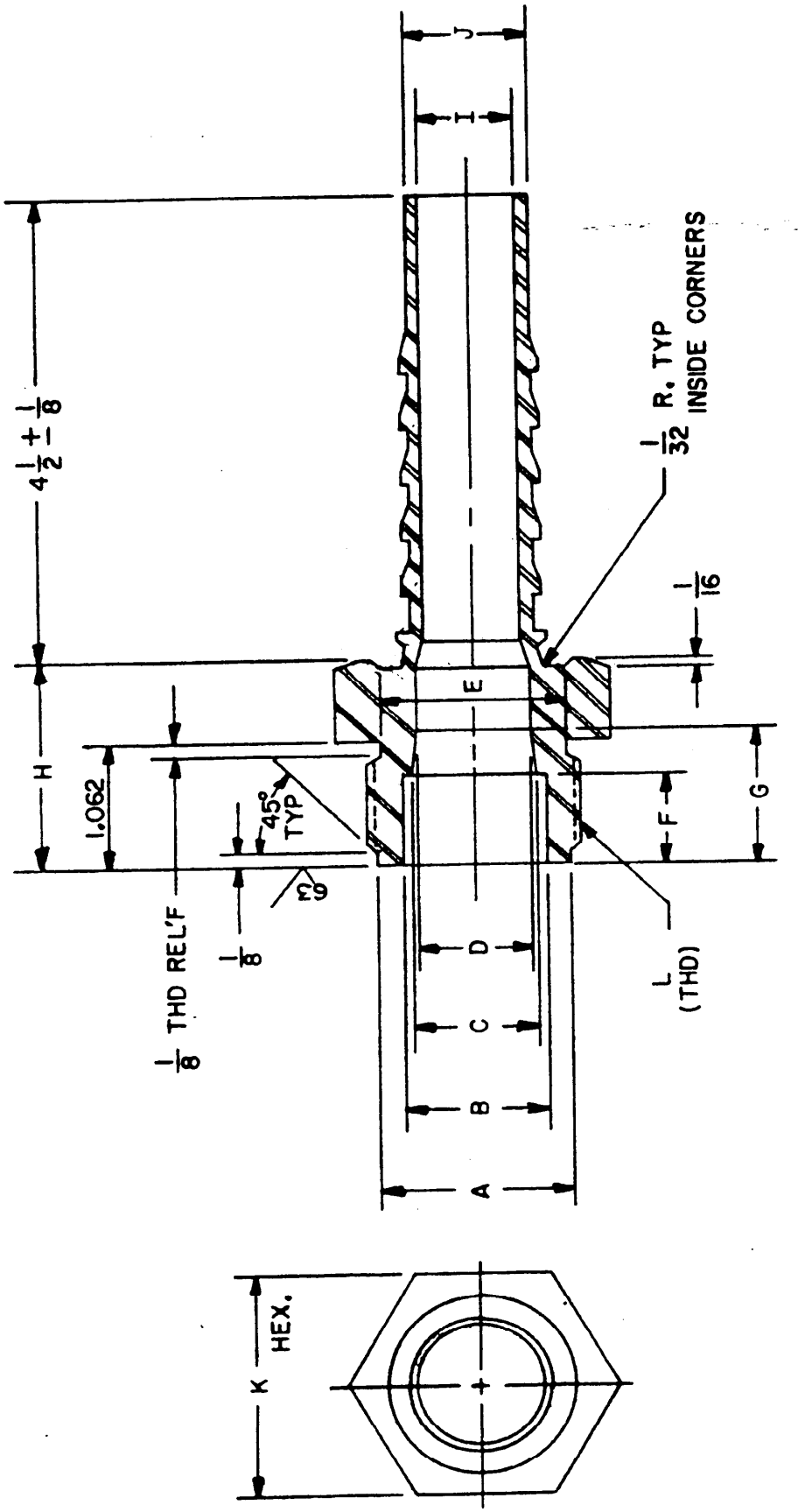
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Hose size (in)	A	B	C	D	E	F	G	H
1	0.060	0.640	0.750	0.844	1.031	1.125	1.315	0.075
1-1/4	1.110	0.870	1.000	1.125	1.281	1.400	1.718	0.937

FIGURE 1. Nose fitting, nipple, female.

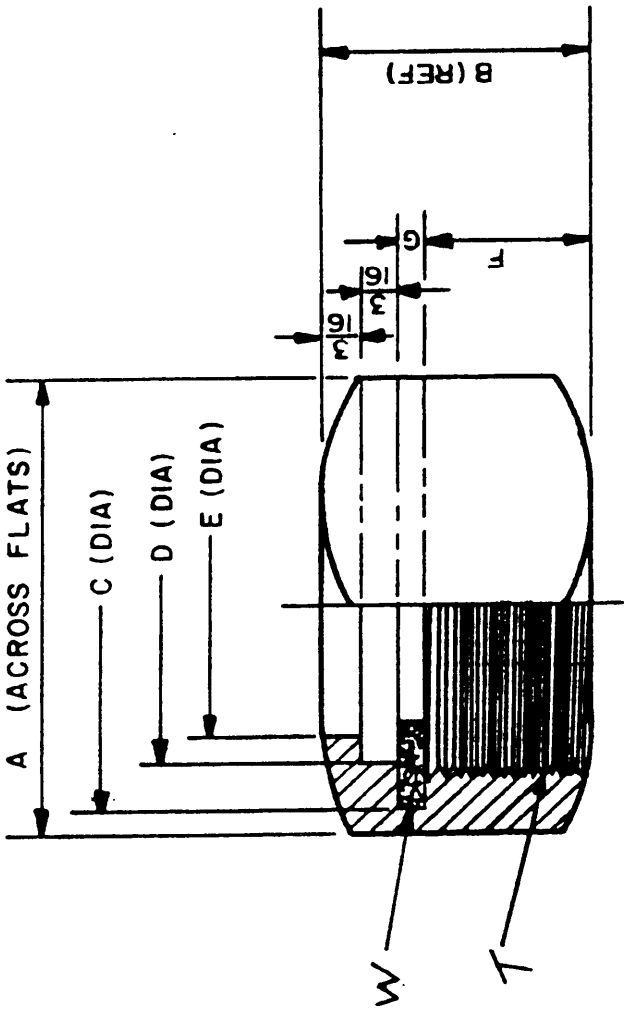
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Hose size (in)	A	B	C	D	E	F	G	H	I	J	K	L (THD)
1	1.522 1.516	1.062	0.844	0.750	1.500	0.750	1.188	1.938	0.640	0.860	2.250	1-1/4 - 11-1/2 NPSH
1-1/4	1.760 1.754	1.312	1.125	1.000	1.688	0.812	1.250	2.000	0.870	1.110	2.500	1-1/2 - 11-1/2 NPSH

FIGURE 2. Hose fitting, coupling, male.

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Hose size (in)	A	B	C	D	E	F	G	T	W (leather washer)
1	2-1/8	1-1/4	1-11/16	1-13/32	1-5/32	3/4	1/8	1-1/4 - 11-1/2 NPSH	1/8 x 1-21/32 od x 1-1/32 id
1-1/4	2-1/2	1-3/8	1-15/16	1-3/4	1-15/32	27/32	5/32	1-1/2 - 11-1/2 NPSH	5/32 x 1-29/32 od x 1-9/32 id

FIGURE 3. Union nut, hose fitting.

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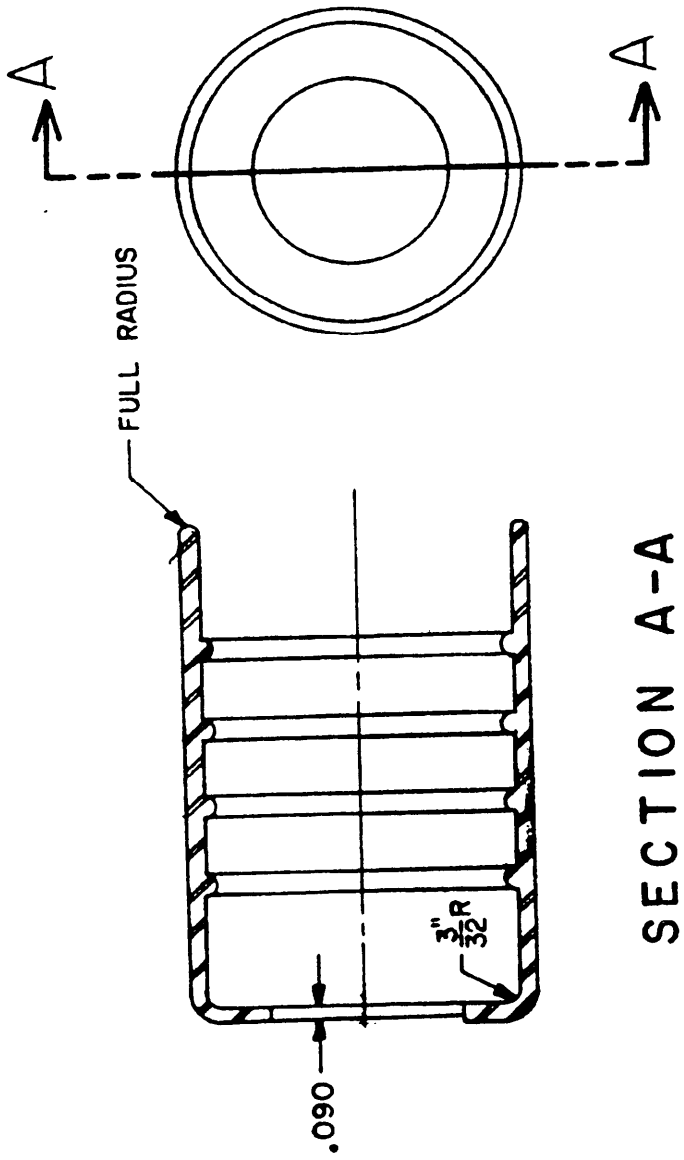
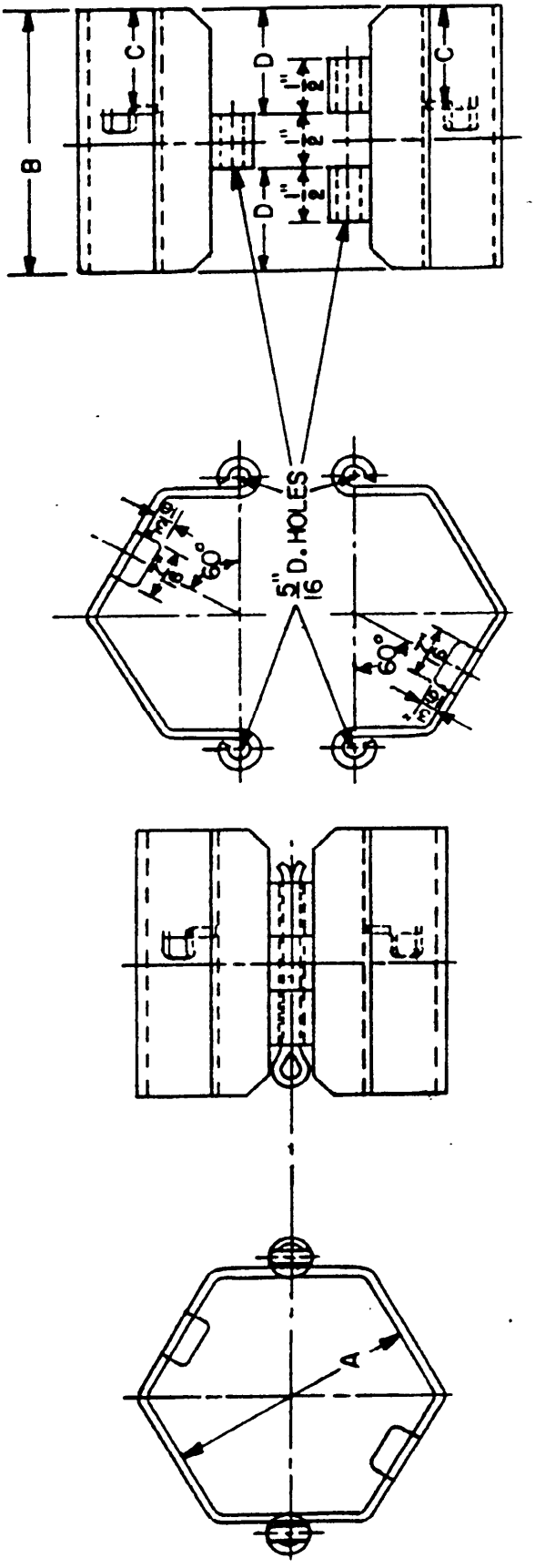


FIGURE 4. Hose fitting collar.

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Hose size (in)	A	B	C	D
1	2-5/32	2-7/16	7/8	31/32
1-1/4	2-17/32	2-1/2	15/16	1

FIGURE 5. Coupling retainer clamp.

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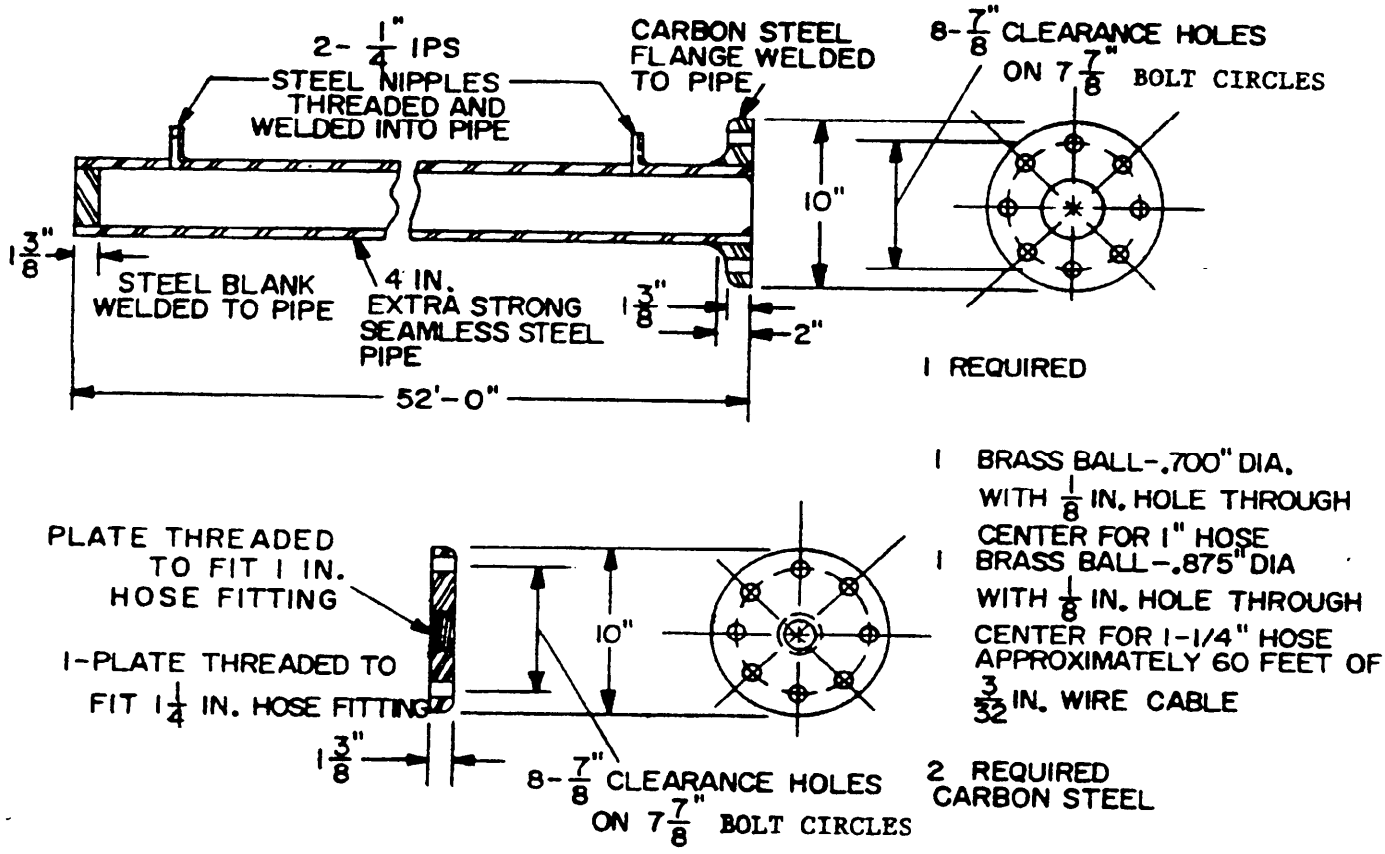


FIGURE 6. Testing chamber.

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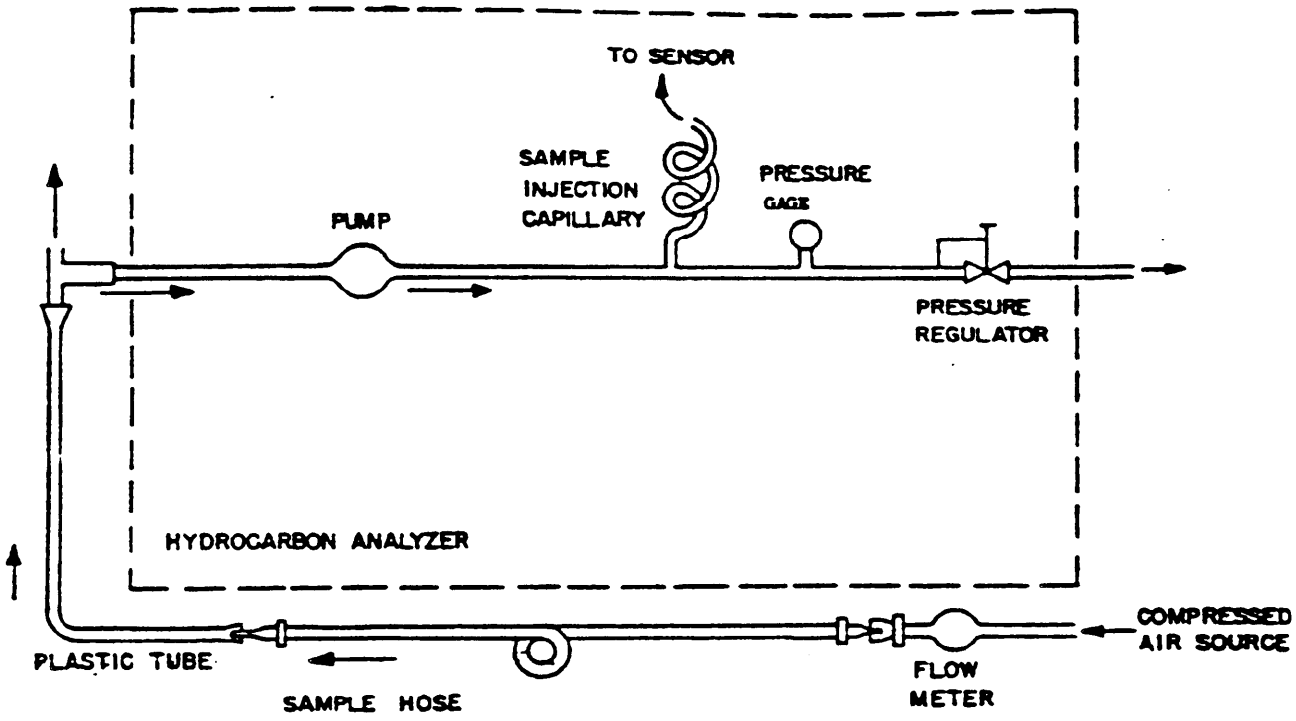


FIGURE 7. Flow apparatus for measuring the concentration of hydrocarbons in a stream of air or other gas after it has passed through a test hose.

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